

Multiple Choice

1. **Answer:** B

Options: A) The resonance pulls Io in different directions and generates heat.; B) It makes the orbit of Io slightly elliptical.; C) It creates a gap with no asteroids between the orbits.; D) It prevents formation of the ring material into other moons.

Note: The 1:2:4 resonance among Jupiter's moons Io, Europa, and Ganymede results in gravitational interactions that cause Io's orbit to become slightly elliptical, influencing its volcanic activity and overall orbital dynamics.

2. **Answer:** D

Options: A) because the Moon has a nearly circular orbit around the Earth; B) because the Moon does not rotate; C) because the other face points toward us only at new moon when we can't see the Moon; D) because the Moon's rotational and orbital periods are equal

Note: The correct answer is based on the phenomenon of synchronous rotation, where the Moon's rotation on its axis takes the same amount of time as its orbit around the Earth, resulting in the same hemisphere of the Moon always facing our planet.

3. **Answer:** B

Options: A) Spontaneous emission from radioactive atoms.; B) The minimization of gravitational potential energy.; C) Thermally induced collisions.; D) Plate tectonics.

Note: Differentiation is driven by the minimization of gravitational potential energy because, in astronomical bodies, denser materials tend to sink towards the center under the influence of gravity, leading to the formation of layered structures based on density.

4. **Answer:** D

Options: A) fast rotation only; B) a rocky mantle only; C) a molten metallic core only; D) both a molten metallic core and reasonably fast rotation

Note: A terrestrial planet requires both a molten metallic core to facilitate the dynamo effect that generates a magnetic field and reasonably fast rotation to enhance this dynamo action, resulting in a strong magnetic field.

5. **Answer:** D

Options: A) hydrogen compounds ($\text{H}_2\text{OCH}_4\text{NH}_3$) light gases (H He) metals rocks; B) hydrogen compounds ($\text{H}_2\text{OCH}_4\text{NH}_3$) light gases (H He) rocks metals; C) light gases (H He) hydrogen compounds ($\text{H}_2\text{OCH}_4\text{NH}_3$) metals rocks; D) light gases (H He) hydrogen compounds ($\text{H}_2\text{OCH}_4\text{NH}_3$) rocks metals

Note: The correct answer reflects the composition of the solar nebula, where light gases such as hydrogen and helium constitute the majority of its mass, followed by hydrogen compounds, then rocks, and finally metals, aligning with the observed abundance of these materials in the early solar system.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The statement is true because Sagittarius A* is the supermassive black hole located at the center of the Milky Way galaxy, as confirmed by various astronomical observations and research.

7. **Answer:** True

Options: A) True; B) False

Note: A larger object, like a baked potato, retains heat longer than a smaller object because it has a lower volume-to-surface area ratio, which reduces the rate of heat loss through its surface.

8. **Answer:** True

Options: A) True; B) False

Note: Radioactive dating of lunar samples, particularly those obtained during the Apollo missions, has revealed ages ranging from approximately 3 to 4.5 billion years, indicating that many lunar rocks formed during the early solar system, which is significantly older than most terrestrial rocks that have been dated to around 4 billion years or younger due to geological processes that recycle Earth's crust.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because during the third quarter phase of the Moon, it rises at midnight, is highest in the sky at sunrise, and sets at noon, positioning it 90 degrees behind the Sun.

10. **Answer:** True

Options: A) True; B) False

Note: Jupiter and other jovian planets possess substantial amounts of non-gaseous matter, including ices and rocky material, which contribute to their overall mass and structure, despite their classification as 'gas giants.'

Long Form Response

11. **Answer:** The Great Red Spot on Jupiter is a colossal anticyclonic storm characterized by its persistent reddish hue, extensive size-approximately 1.3 times the diameter of Earth-and longevity, having existed for at least 350 years. Its formation is attributed to the planet's rapid rotation and complex atmospheric dynamics, which create a stable vortex system that sustains the storm. The Great Red Spot is significant for planetary science as it provides insights into atmospheric phenomena, including the mechanics of long-lived storms, heat transfer in planetary atmospheres, and the dynamics of Jovian weather systems. Studying this iconic feature enhances our understanding of similar atmospheric processes on exoplanets and contributes to the broader field of comparative planetology.

Note: The answer accurately describes the Great Red Spot's characteristics, including its size, longevity, and formation due to Jupiter's atmospheric dynamics, while also emphasizing its significance in advancing our understanding of atmospheric phenomena in planetary science.

12. **Answer:** The distance from Earth to Altair, being one million times that of the Earth to the Sun, has significant implications for our understanding of the scale of the universe. Given that the average distance from Earth to the Sun is approximately 1.5×10^{11} meters, multiplying this value by one million yields a distance of 1.5×10^{17} meters from Earth to Altair. This vast distance underscores the challenges of interstellar travel and the limitations of current technology in exploring such remote celestial bodies. Additionally, this measurement highlights the vastness of our galaxy and the relative proximity of Altair as a bright star in the Aquila constellation, emphasizing the need for advanced observational techniques to study distant stars effectively.

Note: correct because it accurately calculates the distance from Earth to Altair by multiplying the Earth-Sun distance by one million, demonstrating the immense scale of interstellar distances and the practical challenges of space exploration.